Impact of migrant remittances on child health in Burkina Faso

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Abstract

The increase in migrant remittances in recent years in Burkina Faso may contribute to alleviating household liquidity constraints while improving hygiene and health conditions. Therefore, this article aims to analyze the impact of migrant remittances on child health in Burkina Faso. Data from the 2021 EHCVM-2 were used using the linear regression model with an endogenous treatment effect. The results show that migrant remittances lead to a reduction of 2.598 percentage points in the proportion of children aged 0–5 hospitalized for diarrhea, typhoid, stomach problems and malaria, and a reduction of 2.831 percentage points for children aged 6–18. However, the effect appears to be more pronounced among children aged 6 to 18. It is therefore essential that health reforms take into account the magnitude of remittances, while emphasizing improvements in household living and hygiene conditions. In addition, specific health policies for children aged 6 to 18 would be relevant.

JEL Classifications: F24, I12, P36

1. Introduction

The World Health Organization (1948) defines health as a state of complete physical, mental, and social well-being, not merely the absence of disease or infirmity. Health is crucial for human capital formation, and investments in health pose a significant challenge garnering attention in many developing countries. Grossman's (1972) health capital model suggests that individuals are born with a stock of health that diminishes over time but can be replenished through health investment. Weil (2014) emphasizes health's pivotal role in measuring a country's development. Investing in health offers a sustainable opportunity to enhance population productivity and achieve more significant economic development, capable of reducing or eliminating poverty. This aligns with the third Sustainable Development Goal (SDG) of the 2030 agenda, aiming to ensure the health and well-being of all by improving reproductive, maternal, and child health and reducing major communicable, non-communicable, environmental, and mental illnesses.

In sub-Saharan Africa, health, particularly that of children, remains a major public health challenge despite efforts made in recent decades. Globally, sub-Saharan Africa maintains the highest infant mortality rates, with an average under-five mortality rate of 76 deaths per 1,000 live births in 2019, implying that 1 in 13 children dies before reaching the age of 5 (UN IGME, 2020). In Burkina Faso, despite various policies such as the national policy on maternity care subsidies, results regarding progress remain mixed. In 2015, the Global Health Observatory (2019) estimated high maternal and child mortality rates in Burkina Faso, with 371 maternal deaths per 100,000 live births and 61 deaths per 1,000 children under 5.

These slow advancements in health outcomes can be attributed to factors like low public health investments and, more importantly, the financial constraints faced by many households. Financial constraints create an environment conducive to the development and spread of numerous diseases. Removing these financial barriers could offer an interesting solution to bolster household health in Burkina Faso. However, migrant remittances can serve as an ideal solution given their recent increases.

In Sub-Saharan Africa, the World Bank (2023) estimates migrant remittances at nearly $8 billion in 2021, up from $5 billion in 2010, representing 2.6% and 2.2% of the region's Gross Domestic Product (GDP), respectively. In Burkina Faso, total remittances received by households increased from nearly $112 million in 2010 to almost $202 million in 2021, representing 1.19–2.91% of the GDP (World Bank, 2023). Health-wise, these funds can positively impact household health by providing access to higher-quality healthcare. They enable migrants and their families to improve living conditions and hygiene. In addition to easing household budget constraints, remittances can act as insurance against negative economic shocks.

Theoretical frameworks analyzing the impact of migration and remittances on health are grounded in the New Economics of Labor Migration (NELM) theory by Stark and Bloom (1985) and Stark (1991). The NELM theory views an individual's departure as a result of a family or collective decision, not solely an individual one, aimed at maximizing the joint income of a household where costs and returns are shared among all family members (Stark and Bloom, 1985). Additionally, household members migrate not only due to expected wage differentials but also to partially alleviate household liquidity constraints and/or diversify risks in the absence of credit, insurance, and labor markets (Stark, 1978, and Taylor, 2001). The theory suggests that migrant remittances reduce a household's financial constraints, fostering production and investment activities in an imperfect credit market, allowing for increased investment in human and physical capital.

Conceptual frameworks for analyzing healthcare demand include those of Grossman (1972) and more recently Rosenzweig and Schultz (1983), viewing healthcare demand as derived. Health is considered a production factor in human capital, and its improvement is regarded as an investment (Grossman, 1972).

Empirically, the undeniable role of migrant remittances in improving household health conditions has been demonstrated in various studies. Authors such as Kapri and Jha (2020), Suárez and Fontenla (2022), Dulal (2022), Mahmood et al. (2022), and Ramkissoon and Deonanan (2023) support a positive relationship between migrant remittances and household health. They note that remittances increase per capita health expenditures and reduce the prevalence of undernutrition, the depth of food deficit, the prevalence of stunting, and the infant mortality rate. Their results also reveal an increase in children's life expectancy, accompanied by a decrease in mortality rates due to the receipt of remittances. They suggest that improved health conditions, such as the use of sanitation facilities, are a possible channel through which migrant remittances affect diarrheal mortality. Conversely, Wells et al. (2014), Ivlevs et al. (2019), Kakhkharov et al. (2021), and Lei et Desai (2021) find a negative relationship. Their results show that household member migration is associated with increased stress and depression, not compensated by remittances. They also argue that remittances can be unreliable income sources, leaving families in financially precarious situations.

Given this literature review, several studies have analyzed the effect of remittances on health in various countries. However, to our knowledge, empirical evidence on this effect is lacking in the literature for Burkina Faso. Additionally, most research has focused on the health of women and children under 5. This current study aims to extend the findings by analyzing the impact of migrant remittances on health, particularly that of children aged 0 to 18, in Burkina Faso. It seeks to determine the effect of migrant remittances on children's health, approximated by the proportion of children hospitalized for various diseases in the last 12 months within households, an aspect that has been understudied.

The remainder of this research is structured into four sections: The second section discusses the empirical strategy on the relationship between migrant remittances and health. The third section presents the data and summary statistics. Results of econometric estimations are discussed in the fourth section, and the final section concludes the research by drawing key economic policy implications.

2. Methodology

2.1. Modeling the effect of migrant remittances on health

A variety of models are employed in the literature to analyze the impact of migrant remittances on household outcomes. One method for evaluating the counterfactual is the instrumental variable (IV) method, assuming the use of at least one instrument that influences treatment and has no direct effect on the outcome. This method addresses selection bias and endogeneity issues of treatment. However, finding a suitable instrument is often challenging. The matching method proposed by Rubin (1977) estimates the causal effect of treatment by comparing treated and untreated individuals with similar observable characteristics. The Propensity Score Matching method by Rosenbaum and Rubin (1983) assumes that parameters affecting adopters and non-adopters are the same. One limitation of this method is its disregard for unobservable characteristics affecting the participation decision (Adebayo et al., 2018). Additionally, it introduces bias in estimation when unable to compare adopters and non-adopters in a common propensity score framework (Heckman et al., 1998). The two-stage Heckman (1979) model and the Tobit model can be used. However, the Tobit model does not correct biases that may exist in household characteristics. The Endogenous Switching Regression (ESR) model, an improvement over the two-stage Heckman method, also addresses endogeneity and selection bias issues. One limitation is the need for at least one explanatory variable serving as an instrument (Lokshin et al., 2004).

For this research, the analysis method relies on the linear regression model with an endogenous treatment effect. Other authors such as Cameron and Trivedi (2005) and Wooldridge (2010) describe the model as an endogenous treatment effect model and link it to recent work. It differs from matching by propensity scores as it considers biases resulting from observable and unobservable factors. The linear regression model with an endogenous treatment effect is seen as an appropriate procedure for estimating the effect of treatment on an outcome variable. It is a parametric method modeling the link between observable and unobservable variables (Awotide et al., 2015 and Anang et al., 2020).

2.2. Empirical specification of the model

The fundamental idea behind the endogenous treatment linear regression model is that the treatment can be endogenous and distort the estimates of the effects of treatment participation. To estimate the endogenous treatment linear regression model, an equation for the treatment (the probability of receiving remittances) and an outcome equation explaining the proportion of hospitalized children are specified.

The outcome equation for household i is written as follows:

$$Y_i = \alpha X_i + \delta T_i + \mu_i \,(1)$$

Thus, the probability of receiving remittances from migrants (T_i) is assumed to be determined by a latent index (T_i^*), representing the propensity to receive remittances.

$$T_i^* = \beta Z_i + \nu_i \mathrm{with} T_i = \begin{cases} 1 si \beta Z_i + \nu_i > 0 \\ 0 sinon \end{cases} \,(2)$$

X_i and Z_i are the exogenous and observable characteristics explaining the outcome and selection equations, the parameters α and β are the regression coefficients and μ_i and ν_i are the error terms of the two equations, accounting for unobservable that could influence the outcome measure. These terms are assumed to be bivariate normal with mean zero and covariance matrix

$$\begin{pmatrix} \sigma^2 & \rho\sigma \\ \rho\sigma & 1 \end{pmatrix}$$

.

Given that the treatment variable is an endogenous binary variable, the evaluation task involves using observed variables to estimate the regression coefficients α while controlling for selection bias.

The model expressed by equations (1) and (2) is a regime-switching regression. By substituting Eq. (2) into Eq. (1), two distinct outcome regression equations are obtained.

$$Y_i = \alpha_i X_i + \delta + \mu_i lorsque T_i = 1 \,(3)$$

$$Y_i = \alpha_i X_i + \mu_i lorsque T_i = 0 \,(4)$$

The model facilitates the determination of the Average Treatment Effect (ATE) and the Average Treatment Effect on Treated Households (ATT). However, if the treatment indicator variable is not interacted with any of the covariates of the outcome variable, the model directly estimates the Average Treatment Effect on Treated Households (ATT).

3. Data and descriptive statistics

3.1. Data

The data utilized in this study originate from the second Harmonized Household Living Conditions Survey (EHCVM-2). This survey was conducted across the 8 member countries of the West African Economic and Monetary Union (WAEMU). The primary objective of the EHCVM is to generate indicators for monitoring and evaluating household poverty and living conditions. Specifically, it aims to produce indicators related to poverty, education, health, employment, non-agricultural family businesses, consumption, food and non-food expenses, income, savings and credit, housing, water and sanitation, shocks, and survival strategies, as well as agriculture and livestock.

In Burkina Faso, the study was conducted by the National Institute of Statistics and Demography (INSD) between 2021–2022. The representativeness of the collected data pertains to the residential areas (urban and rural) and the 13 administrative regions of the country. Data collection activities in the field took place in two waves, each accounting for half of the sample. The first wave occurred in the months of September, October, and November 2021, and the second in the months of April, May, and June 2022.

In this study, our focus is on children aged 0 to 18 and the characteristics of their households in rural and urban areas of Burkina Faso. Thus, our unit of observation for this research is the household. After processing information related to sick children who underwent hospitalizations in the last 12 months in each household and the characteristics of households, our sample consists of 7130 households.

The dependent or outcome variable is captured by the proportion of hospitalized children, measuring the health status of children within households. It represents the share of children in the household who experienced health issues leading to hospitalizations in the 12 months preceding the survey (Kan, 2021). As health issues, we refer to diseases that can be linked to poverty and hygiene problems such as diarrhea, typhoid fever, stomach issues, and malaria (Ramkissoon and Deonanan, 2023). Additionally, we focused more on children aged 0 to 18, categorizing them into two groups: 0 to 5 years and 6 to 18 years, to better capture the distinct impact of remittances for each age bracket. Indeed, children's health is closely linked to their well-being and optimal development. Healthy children are more likely to achieve better academic results, engage in social activities and cultivate their skills and talents, thereby promoting personal fulfillment and happiness. In addition, prioritizing children's health strengthens both individual and collective capacities within society, while contributing to a country's economic growth.

The variable of interest is the receipt of migrant remittances. It is a binary variable taking the value of 1 if the household receives remittances and 0 if not. The receipt of migrant remittances is expected to influence the proportion of hospitalized children within households (Kapri and Jha, 2020; Suárez and Fontenla, 2022; Ramkissoon and Deonanan, 2023).

In addition to these variables, existing literature on previous studies informs us about independent variables related to the household that may affect children's health. The different variables and their expected effects are documented in the following table:

Table 1
Variables and their expected impact on children's health

Variable	Definition	Expected effect
	Dependent variables	
Y1	Proportion of children hospitalized from 0 to 5 years	
Y2	Proportion of children hospitalized from 6 to 18 years	
	Treatment variable	
T	1 if the household receives remittances and 0 otherwise	-
	Socioeconomic and demographic variables	
Age-CM	Age in years of the household head	+/-
Gender-CM	1 if the household head is male and 0 otherwise	+
Marital status-CM	1 if the household head is married and 0 otherwise	-
Main activity-CM	1 if the main occupation of the household head is agriculture and 0 otherwise	+
Income	Household income in thousands of FCFA	-
Access to credit	1 if the household has access to credit and 0 otherwise	-
Dependency ratio	Ratio of inactive to active members in the household	+
Literacy-CM	1 if the household head is literate and 0 otherwise	-
Formal education-CM	0 if the household head has no education	Reference
	1 if the household head has primary education	-
	2 if the household head has secondary education	-
Access to toilets	1 if the household has a latrine and 0 otherwise	-
Access to electricity	1 if the household has a television and 0 otherwise	-
Healthy water disposal	1 if the household has a proper water evacuation system and 0 otherwise	-
Notes: CM = Head of household		
Source: EHCVM-2 data, 2021		

Variable	Definition	Expected effect
	Dependent variables	
Healthy waste disposal	1 if the household has a proper waste disposal system and 0 otherwise	-
Home ownership	1 if the household owns the residence and 0 otherwise	-
Residential area	1 if the household resides in an urban area and 0 otherwise	-
Number of inactive	The number of individuals who are not active in the labor market	+
Notes: CM = Head of household		
Source: EHCVM-2 data, 2021		

3.2. Descriptive statistics

Table 2 shows the descriptive statistics of the different variables used in the analysis. According to this table, 32.4% of households receive remittances and only 43.7% live in urban areas. Furthermore, the sample selected for analysis shows that the proportion of hospitalized children aged 0 to 5 years is 3.49%, compared to 3.21% for children aged 6 to 18 years. However, the proportion of hospitalized children appears to be higher in households that do not receive migrant remittances than in those that do (3.5% versus 3.32% for children aged 0–5, and 3.28% versus 3.06% for children aged 6–18). This confirms the important role of migrant remittances in improving the health status of children in recipient households.

Table 2
:Mean and standard deviation of utilized variables

Variable	Average	Reception	No reception	Difference
Remittances of funds	0.324 (0.468)			
Children 0–5 years	0.0349 (16.210)	0.0332 (15.508)	0.035(16.512)	0.0025
Children aged 6–18	0.0321(13.298)	0.0306 (11.496)	0.0328 (14.09)	0.0022
Age-CM	0.469 (0.146)	0.514 (0.173)	0.447 (0.131)	-0.067***
Gender-CM	0.849 (0.357)	0.752 (0.431)	0.895 (0.305)	0.143***
Marital status-CM	0.890 (0.312)	0.836 (0.369)	0.916 (0.276)	0.079***
Main activity-CM	0.436 (0.496)	0.447 (0.497)	0.431 (0.495)	-0.015
Income	2409.055(1684.8)	2537.68(1790.3)	2347.252(1628.3)	-190.4***
Access to credit	0.042 (0.202)	0.045 (0.209)	0.041 (0.199)	-0.004
Dependency ratio	1.573 (1.164)	1.656 (1.502)	1.532 (0.958)	-0.124***
Literacy-CM	0.137 (0.344)	0.130 (0.336)	0.140 (0.347)	0.010
Formal education-CM				
None	0.698 (0.459)	0.714 (0.451)	0.690 (0.462)	-0.024
Primary	0.157 (0.364)	0.157 (0.364)	0.157 (0.364)	0
Secondary	0.144 (0.351)	0.127 (0.333)	0.152 (0.359)	0.025
Access to toilets	0.172 (0.378)	0.199 (0.399)	0.159 (0.366)	-0.039***
Access to electricity	0.684 (0.464)	0.687 (0.463)	0.682 (0.465)	-0.005
Healthy water disposal	0.979 (0.142)	0.989 (0.101)	0.974 (0.158)	-0.015***
Healthy waste disposal	0.188 (0.390)	0.187 (0.390)	0.188(0.391)	0.001
Home ownership	0.796 (0.402)	0.816 (0.386)	0.787 (0.409)	-0.029***
Residential area	0.437 (0.496)	0.436 (0.495)	0.437 (0.496)	0.001

Notes: CM= Head of household

Source: EHCVM-2 data, 2021

******p<0.01, **p<0.05 and *p<0.1 are significance at 1%, 5% and 10% respectively.***

Standard deviations are shown in brackets.

Remittance-receiving households have a relatively higher average income (XOF 2537.68 thousand) than non-recipient households (XOF 2347.252 thousand). In addition, remittance-receiving households have better access to credit and are generally better educated.

In terms of improving living conditions and hygiene, remittance-receiving households have relatively higher average rates of access to electricity, toilets, and housing. They also have better sanitation than households that do not receive remittances. These variables are likely to influence the proportion of children hospitalized, and this is confirmed by econometric analysis.

The mean difference test indicates a significant difference in the age of the household head, gender, marital status, income, dependency ratio, and education level between migrant remittance recipient and non-recipient households. The same is true for access to toilets, housing, and the process of healthy water drainage.Table 2 :Mean and standard deviation of utilized variables

Table 3 :Results of the Effect of Migrant Fund Transfers on the Proportion of Hospitalized Children, Ages 0 to 5

Variables	Probability of reception		Proportion of children hospitalized	
Age-CM	-3.353***	(0.751)	-24.890***	(9.109)
Age squared-CM	4.633 ***	(0.724)	25.713 ***	(9.038)
Gender-CM	-0.993***	(0.070)	-1.553	(0.999)
Marital status (ref : married)	0.226 ***	(0.086)	1.104	(1.087)
Literacy-CM	-0.034	(0.057)	1.980 ***	(0.669)
Formal education-CM (ref: none)				
Primary	0.099 *	(0.055)	1.157 *	(0.653)
Secondary	0.115 *	(0.063)	0.380	(0.7487)
Main activity-CM	0.032	(0.040)	1.059 **	(0.522)
Income	-0.00002	(0.00002)	0.0004	(0.0003)
Income squared	-8.84E-07	(2.51E-06)	-0.00004	(0.00003)
Dependency ratio	0.026	(0.019)	0.208	(0.215)
Access to credit	0.125	(0.094)	3.866***	(1.116)
Number of inactive	0.163***	(0.011)		
Access to electricity			-2.289***	(0.517)
Access to toilets			0.243	(0.626)
Healthy waste disposal			-0.075	(0.613)
Healthy water disposal			-8.411***	(1.650)
Home ownership			0.551	(0.590)
Residential area			0.692	(0.549)
Remittances of funds			-2.598 **	(1.315)
Constant	0.023	(0.195)	17.045 ***	(2.981)
Athrho	0.092**	(0.046)		
Lnsigma	2.780 ***	(0.009)		
Rho	0.092	(0.046)		
Sigma	16.125	(0.160)		
Lambda	1.494	(0.749)		
	Chi2 (19) = 82.11			

	Prob > chi2 = 0.000
Number of observations	5315
Wald test of indep. eqns. (rho = 0): chi2(1) = 3.96 Prob > chi2 = 0.046	

Notes: CM= Head of household

Source: EHCVM-2 data, 2021

******p<0.01, **p<0.05 and *p<0.1 are respectively the significance at 1%, 5% and 10%.***

Table 4: Results of the effect of migrant remittances on the proportion of children hospitalized, ages 6 to 18

Variables	Probability of reception		Proportion of children hospitalized	
Age-CM	-3.724 ***	(0.714)	1.517	(7.197)
Age squared-CM	5.311 ***	(0.678)	-1.755	(7.029)
Gender-CM	-0.813 ***	(0.059)	-3.606 ***	(0.672)
Marital status (ref : married)	0.083	(0.067)	0.082	(0.690)
Literacy-CM	-0.012	(0.052)	0.117	(0.508)
Formal education-CM (ref: none)				
Primary	0.202 ***	(0.050)	1.388***	(0.503)
Secondary	0.139 **	(0.059)	-0.003	(0.588)
Main activity-CM	0.079 **	(0.037)	-0.025	(0.399)
Income	-7.13E-06	(0.00002)	0.001 ***	(0.0002)
Income squared	-1.58E-06	2.30E-06	-0.00007 ***	(0.00002)
Dependency ratio	0.070 ***	(0.017)	-0.205	(0.151)
Access to credit	0.114	(0.086)	-0.361	(0.845)
Number of inactive	0.148***	(0.010)		
Access to electricity			-0.341	(0.387)
Access to toilets			-0.853 *	(0.474)
Healthy waste disposal			0.171	(0.461)
Healthy water disposal			-0.776	(1.167)
Home ownership			-0.525	(0.455)
Residential area			-0.437	(0.417)
Remittances of funds			-2.831 ***	(0.977)
Constant	-0.089	(0.192)	6.537 ***	(2.315)
Athrho	0.110***	(0.042)		
Lnsigma	2.585***	(0.009)		
Rho	0.110	(0.041)		
Sigma	13.267	(0.123)		

Lambda	1.465	(0.556)
	Chi2 (19) = 75.03	
	Prob > chi2 = 0.000	
Number of observations	6272	
Wald test of indep. eqns. (rho = 0): chi2(1) = 6.91 Prob > chi2 = 0.008		

Notes: CM= Head of household

Source: EHCVM-2 data, 2021

******p<0.01, **p<0.05 and *p<0.1 are respectively the significance at 1%, 5% and 10%.***

Statistics in brackets

4. Results of estimates and discussions

This section is dedicated to presenting the results of the effect of remittances on the proportion of hospitalized children aged 0 to 5 and 6 to 18 within households.

4.1. Validity tests of results

The validity of the endogenous treatment regression model is based on an LR test that examines the independence of the equations for each regression. It is assessed using a Chi-square test that examines the significance of the correlation coefficients (rho) linking the treatment and outcome equations. Tables 3 and 4 show that the rho coefficients are statistically significant and positive for both regressions, at 5% for the proportion of children aged 0 to 5 hospitalized and 1% for those aged 6 to 18. The independence test for each regression equation also indicates that we can reject the null hypothesis of no correlation between treatment assignment errors and outcome assignment errors for the control and treatment groups. Moreover, the significance of rho suggests the presence of unobservable biases that affect both the probability of receiving transfers and the proportion of hospitalized children within households. The use of the endogenous treatment effect model is therefore justified to correct for these biases.

4.2. Results of the endogenous treatment regression model

4.2.1. Determinants of the probability of receiving migrant remittances

Following the estimation of the selection equations, the results in Tables 3 and 4 show that the threshold effect of the age of the household head positively influences the probability of receiving migrant remittances. This suggests that households headed by older individuals are more likely to receive remittances than those headed by young individuals of working age. This may be explained by the vulnerability associated with age-related disabilities. These findings are consistent with Akim and Robilliard (2019).

The results show a negative relationship between the gender of the household head and the receipt of migrant remittances. A possible explanation for this finding is that in African societies, labor migration is predominantly a male-dominated activity, and women are potential recipients of these remittances. As heads of households, women play a crucial role as recipients and managers of remittances. In addition, male household heads are less vulnerable and have greater physical capabilities than women to engage in various income-generating activities. This finding confirms Kebe and Charbit (2007).

Marital status positively affects the likelihood of receiving migrant remittances. When the household head is married, there are more responsibilities, and consumption, education, and health expenditures become more important because there is a higher likelihood of having more family members to care for.

Household heads with formal education (primary or secondary) are more likely to receive migrant remittances than those without formal education. The educational level of the household can be seen as an indicator of wealth, hence the observed positive correlation with the probability of receiving migrant remittances. This finding is similar to that of Andersson (2014).

The dependency ratio and the number of inactive members positively influence the probability of receiving migrant remittances. A higher number of inactive individuals in a household lead to significant expenditures on food, hygiene and health, as this segment of the population is the most vulnerable. Therefore, there is an increase in migrant remittances. This result is in line with Mahmood et al. (2022).

4.2.2. Effect of migrant remittances on the proportion of hospitalized children

Tables 3 and 4 show that the estimated Average Treatment Effects (ATE) due to receiving migrant remittances are − 2.598 and − 2.831, respectively, for the proportion of children hospitalized aged 0 to 5 and aged 6 to 18. The estimated ATEs in these regressions also correspond to the Average Treatment Effect on the Treated (ATT). They represent the average effect of receiving migrant remittances on the proportion of children hospitalized for specific illnesses such as diarrhea, typhoid fever, stomach problems, and malaria, which are generally associated with poverty and poor living and hygiene conditions. The results indicate that receiving migrant remittances decreases the proportion of hospitalized children aged 0 to 5 by 2.598 percentage points and the proportion of hospitalized children aged 6 to 18 by 2.831 percentage points. These findings reveal the significant role of migrant remittances in improving household living conditions through poverty reduction, health improvement, and contribution to the country's growth. Furthermore, these funds can serve as a mechanism for health shocks. They can act as social protection or a safety net for the poor in the absence of a formal social insurance program. They enable households to improve their living conditions (construction of housing, latrines, access to water, electricity, etc.), as the living environment of households or children influences their health and chances of leading a fulfilling life.

Moreover, migrant remittances can provide beneficiary households with economic capital, enabling them to engage in small businesses and invest in agriculture, livestock, and land, allowing them to cope with health expenses. In the presence of remittances, child labor may be reduced. This result is consistent with Suárez and Fontenla (2022), Dulal (2022), Mahmood et al. (2022), and Ramkissoon and Deonanan (2023).

Additionally, the results show that the effect of migrant remittances on the proportion of hospitalized children aged 6 to 18 appears to be more significant compared to that of children aged 0 to 5 (i.e., ATE = -2.831 versus ATE = -2.598). This can be explained by the free healthcare policy implemented in Burkina Faso since 2016 in public health facilities for children aged 0 to 5. Established to eliminate financial barriers to health services, through this policy, the Government of Burkina Faso covers the entire cost of a defined set of health services for mothers, newborns, and children. This set of pediatric services often focuses on nutrition while monitoring the growth and development of the child. They play an important role in prevention and detection, aiming to diagnose and treat pathologies that can affect their health. These services help prevent certain diseases such as diarrhea, typhoid fever, and stomach problems compared to children aged 6 to 18. These results are in line with Tiehi and Coulibaly (2022), who estimate that government expenditures through environmental and social spending reduce disparities in maternal and child mortality rates in UEMOA countries. Another possible explanation is that children aged 0 to 5 benefit more from parental supervision and are often exempt from child labor (domestic work, livestock, agriculture, etc.) compared to those aged 6 and older.

As for the heterogeneity effect for both age groups, it shows that the proportion of hospitalized children in recipient households of remittances is lower compared to the proportion in non-recipient households. This difference is explained by the presence of significant sources of heterogeneity.

4.2.3. Effect of other explanatory variables on the proportion of hospitalized children

Household income has a threshold effect that negatively affects the proportion of hospitalized children aged 6 to 18 within households. This is due to the fact that in impoverished households, increases in income do not translate directly into improved hygiene and health conditions for household members. Additional income at this level may be used to purchase agricultural equipment or for other purposes rather than to improve household living and hygiene conditions. Thus, investments in health through improved living conditions are observed only after a certain level of wealth is reached. This finding confirms that of N'Guessan (2008).

The results indicate a negative relationship between the gender of the head of the household and the proportion of hospitalized children aged 6 to 18. One possible explanation for this finding is that men predominate in household leadership and decision-making roles and are more likely to adopt better health behaviors than their female counterparts. In addition, male-headed households may be more likely to face financial constraints that limit their access to health care for their children. This finding is consistent with that of Cisse (2011).

The age of the household head is associated with a threshold effect that positively affects the proportion of hospitalized children aged 0–5. This may be explained by the fact that as household heads age, they may also be more prone to developing health problems, financial constraints, or other disabilities that could affect their ability to adequately care for their children. In terms of access to credit, it increases the proportion of hospitalized children aged 0–5. A possible explanation for this finding is that credit may be used for purposes other than accessing quality health care or improving household hygiene and living conditions.

Household access to toilets, electricity, and a good water drainage system negatively affects the proportion of children hospitalized. Access to toilets, electricity and a good water drainage system are all essential elements in maintaining a safe and hygienic home environment, which can reduce the risk of infection and disease, including diarrheal, respiratory and parasitic infections in children. Thus, improving living and hygiene conditions can reduce the incidence of communicable and noncommunicable diseases and improve the overall well-being of household members. This finding corroborates that of Cingolani et al. (2015).

The literacy and formal (primary) education level of the head of the household have a positive effect on the proportion of children hospitalized. This can be explained by the fact that educated household heads tend to rely on self-medication to save time, money, etc. In addition, they are often influenced by their knowledge and advertisements, previous prescriptions, and suggestions from family members or friends. As a result, these actions can aggravate the disease and lead to hospitalization.

The main occupation of the head of the household has a positive effect on the proportion of hospitalized children. This can be explained by the fact that agriculture in Burkina Faso is still an embryonic activity, with relatively low agricultural incomes, which can lead to limited access to resources and services necessary to ensure the health and well-being of the household. In addition, agricultural households are often exposed to risks such as the use of pesticides, strenuous physical labor, and exposure to extreme environmental conditions, which can increase the risk of illness and accidents among children. They also often live in precarious conditions in terms of housing, sanitation and access to clean water, which can increase the risk of disease.

Table 3
:Results of the Effect of Migrant Fund Transfers on the Proportion of Hospitalized Children, Ages 0 to 5

Variables	**Probability of reception**		**Proportion of children hospitalized**	
Age-CM	-3.353***	(0.751)	-24.890***	(9.109)
Age squared-CM	4.633 ***	(0.724)	25.713 ***	(9.038)
Gender-CM	-0.993***	(0.070)	-1.553	(0.999)
Marital status (ref : married)	0.226 ***	(0.086)	1.104	(1.087)
Literacy-CM	-0.034	(0.057)	1.980 ***	(0.669)
Formal education-CM (ref: none)				
Primary	0.099 *	(0.055)	1.157 *	(0.653)
Secondary	0.115 *	(0.063)	0.380	(0.7487)
Main activity-CM	0.032	(0.040)	1.059 **	(0.522)
Income	-0.00002	(0.00002)	0.0004	(0.0003)
Income squared	-8.84E-07	(2.51E-06)	-0.00004	(0.00003)
Dependency ratio	0.026	(0.019)	0.208	(0.215)
Access to credit	0.125	(0.094)	3.866***	(1.116)
Number of inactive	0.163***	(0.011)		
Access to electricity			-2.289***	(0.517)
Access to toilets			0.243	(0.626)
Healthy waste disposal			-0.075	(0.613)
Healthy water disposal			-8.411***	(1.650)
Home ownership			0.551	(0.590)
Residential area			0.692	(0.549)
Remittances of funds			-2.598 **	(1.315)
Constant	0.023	(0.195)	17.045 ***	(2.981)
Athrho	0.092**	(0.046)		
Lnsigma	2.780 ***	(0.009)		
Notes: CM = Head of household				
Source: EHCVM-2 data, 2021				

Variables	Probability of reception		Proportion of children hospitalized
Rho	0.092	(0.046)	
Sigma	16.125	(0.160)	
Lambda	1.494	(0.749)	
	Chi2 (19) = 82.11		
	Prob > chi2 = 0.000		
Number of observations	5315		
Wald test of indep. eqns. (rho = 0): chi2(1) = 3.96 Prob > chi2 = 0.046			
Notes: CM = Head of household			
Source: EHCVM-2 data, 2021			

*****p < 0.01, **p < 0.05 and *p < 0.1 are respectively the significance at 1%, 5% and 10%.**

Table 4
Results of the effect of migrant remittances on the proportion of children hospitalized, ages 6 to 18

Variables	Probability of reception		Proportion of children hospitalized	
Age-CM	-3.724 ***	(0.714)	1.517	(7.197)
Age squared-CM	5.311 ***	(0.678)	-1.755	(7.029)
Gender-CM	-0.813 ***	(0.059)	-3.606 ***	(0.672)
Marital status (ref : married)	0.083	(0.067)	0.082	(0.690)
Literacy-CM	-0.012	(0.052)	0.117	(0.508)
Formal education-CM (ref: none)				
Primary	0.202 ***	(0.050)	1.388***	(0.503)
Secondary	0.139 **	(0.059)	-0.003	(0.588)
Main activity-CM	0.079 **	(0.037)	-0.025	(0.399)
Income	-7.13E-06	(0.00002)	0.001 ***	(0.0002)
Income squared	-1.58E-06	2.30E-06	-0.00007 ***	(0.00002)
Dependency ratio	0.070 ***	(0.017)	-0.205	(0.151)
Access to credit	0.114	(0.086)	-0.361	(0.845)
Number of inactive	0.148***	(0.010)		
Access to electricity			-0.341	(0.387)
Access to toilets			-0.853 *	(0.474)
Healthy waste disposal			0.171	(0.461)
Healthy water disposal			-0.776	(1.167)
Home ownership			-0.525	(0.455)
Residential area			-0.437	(0.417)
Remittances of funds			-2.831 ***	(0.977)
Constant	-0.089	(0.192)	6.537 ***	(2.315)
Athrho	0.110***	(0.042)		
Notes: CM = Head of household				
Source: EHCVM-2 data, 2021				

Variables	Probability of reception		Proportion of children hospitalized
Lnsigma	2.585***	(0.009)	
Rho	0.110	(0.041)	
Sigma	13.267	(0.123)	
Lambda	1.465	(0.556)	
	Chi2 (19) = 75.03		
	Prob > chi2 = 0.000		
Number of observations	6272		
Wald test of indep. eqns. (rho = 0): chi2(1) = 6.91 Prob > chi2 = 0.008			
Notes: CM = Head of household			
Source: EHCVM-2 data, 2021			

*****p < 0.01, **p < 0.05 and *p < 0.1 are respectively the significance at 1%, 5% and 10%.**

Statistics in brackets

5. Conclusion and implications

Investments in the healthcare sector remain consistently low in developing countries, including Burkina Faso. This inadequacy in healthcare investments is partially attributed to financial constraints, where migrant remittances can offer a viable solution. Consequently, this research aimed to analyze the impact of remittances on child health in Burkina Faso, utilizing data from the 2021 EHCVM-2 through the endogenous treatment regression model.

According to the results of the first regression (selection equations), formal education level, marital status, primary occupation, age threshold effect of the household head, dependency ratio, and the number of inactive members positively influence the probability of receiving migrant remittances. This implies that an increase in any of these factors raises the likelihood of receiving migrant remittances.

The results of the second regression (outcome equations) reveal that the gender of the household head, income threshold effect, access to electricity, toilets, proper water drainage, and receiving remittances decrease the proportion of hospitalized children in a household. This suggests that an increase in any of these factors improves the reduction in the proportion of hospitalized children.

The estimated Average Treatment Effects (ATE) demonstrate that receiving migrant remittances decreases the proportion of children aged 0 to 5 hospitalized due to diarrhea, typhoid fever, stomach problems, and malaria by 2.598 percentage points. Similarly, the proportion of children aged 6 to 18 hospitalized is reduced by 2.831 percentage points. However, the effect of remittances on the proportion of hospitalized children aged 6 to 18 appears to be more substantial compared to those aged 0 to 5. This study revealed that receiving migrant remittances reduces the proportion of hospitalized children within a household after controlling for observable and unobservable factors. These findings support the notion that migrant remittances can serve as a channel for improving household health.

Based on these results, several implications for economic policy can be drawn in terms of strategies to enhance child health in Burkina Faso. Given the significant role played by remittances in improving household health, the primary implication is to implement incentive measures to increase the volume of remittances and channel them towards the countries of origin. Furthermore, it is recommended to strengthen the improvement of living conditions and hygiene in households, such as access to toilets, electricity, and proper water drainage. Additionally, policy formulation focused on the health of children aged 6 to 18 would be advisable.

Declarations

Conflict of interest statement

The authors declare no conflicts of interest.

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